Mock exam 2: answers

Section A: Multiple choice questions

1 [1] medium

A bar magnet is broken into three pieces. How many poles do the three pieces have in all?

(a) 2 **(b)** 3

(c) 4 **(d)** 6

Answer (d) 6

Every piece is a complete magnet with a North pole and a South pole. Three pieces have 3 x 2 = 6 poles.

Read in the chapter: 4.2 Poles of Magnet

Back to question 1

2 [1] medium

An iron bar is hung freely from a thread and turned gently several times. Each time, it will

(a) rest along the north-south direction **(b)** rest along the east-west direction

(c) rest along any direction **(d)** keep on rotating

Answer (c) rest along any direction

Only a magnet rests along north-south every time. An iron bar is not a magnet, so it can rest in any direction.

Read in the chapter: 4.3 Finding Directions

Back to question 2

3 [1] medium

The North pole of a bar magnet is slowly brought near the North pole of a compass needle. The North pole of the needle

(a) moves closer to the magnet

(b) moves away from the magnet

(c) does not move

(d) starts spinning

Answer (b) moves away from the magnet

The compass needle is also a magnet. Like poles repel, so its North pole moves away.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 3

4 [1] medium

On some magnets the poles are not marked N and S. On such a magnet, the North pole may be shown by

(a) a white dot

(b) blue paint

(c) a black line

(d) a small hole

Answer (a) a white dot

The North pole may be shown by a white dot, or painted red. Blue paint shows the South pole.

Read in the chapter: 4.5 Fun with Magnets

Back to question 4

5 [1] hard

Assertion (A): An iron bar brought near a compass needle cannot push either end of the needle away.
Reason (R): Repulsion takes place only between the like poles of two magnets.

(a) Both A and R are true, and R explains A.

(b) Both A and R are true, but R does not explain A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer (a) Both A and R are true, and R explains A.

An iron bar is not a magnet and has no poles. It can only be attracted, so it cannot repel the needle.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 5

6 [1] medium

A strong magnet is brought near a coin. The coin is not attracted at all. Which conclusion is correct?

(a) The coin is made of a non-magnetic material.

(b) The coin is made of iron.

(c) The magnet has lost one of its poles.

(d) The coin is a magnet.

Answer (a) The coin is made of a non-magnetic material.

A material that is not attracted towards a magnet is non-magnetic. Iron would have been attracted.

Read in the chapter: 4.1 Magnetic and Non-magnetic Materials

Back to question 6

Section B: Very short answer questions

7 [2] medium

A steel paper clip lies at the bottom of a plastic bottle filled with water. How can you take it out without wetting your hand? Why does this work?

Answer Hold a magnet against the outside of the bottle near the clip and slide it up to the mouth. It works because the magnetic effect acts through plastic and water.

1. The clip is attracted by the magnet and follows it. (1 mark)
2. Plastic is a non-magnetic material, and the magnetic effect can act through it. (1 mark)

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 7

8 [2] hard

Sana dips a U-shaped magnet into a heap of iron filings and lifts it. Where will most of the filings stick? What does this tell her about the magnet?

Answer Most filings stick at the two ends (tips) of the U. This tells her that the poles of the U-shaped magnet are at its two ends.

1. Iron filings stick mostly to the poles of a magnet, whatever its shape. (1 mark)
2. So the places with the most filings are the two poles. (1 mark)

Read in the chapter: 4.2 Poles of Magnet

Back to question 8

Section C: Short answer questions

9 [3] hard

Dev has a bar magnet with marked poles and two unmarked metal bars, P and Q.
One end of bar P is attracted by the North pole of the magnet and also by its South pole.
One end of bar Q is attracted by the North pole of the magnet but repelled by its South pole.
(a) Which bar is a magnet? (b) Which pole is at that end of the bar? (c) What is the other bar?

Answer (a) Bar Q is a magnet. (b) That end of Q is a South pole. (c) Bar P is a piece of a magnetic material such as iron, not a magnet.

1. Only a magnet can be repelled by another magnet. Q is repelled, so Q is a magnet.
2. The end of Q is repelled by the South pole, so it is a like pole. It is a South pole.
3. P is attracted by both poles and never repelled. So P is not a magnet, but it is made of a magnetic material.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 9

10 [3] medium

(a) Why can a freely suspended magnet be used to find directions?
(b) Why does it behave in this way?
(c) Why is one end of a compass needle usually painted red?

Answer (a) It always rests along the north-south direction. (b) Because the Earth behaves like a giant magnet. (c) The red end shows the end that rests towards the North.

1. A freely suspended magnet comes to rest along north-south every time, so it shows us these directions.
2. Our Earth itself behaves like a giant magnet.
3. The needle is a magnet. Painting the north-pointing end red tells us which way is North.

Read in the chapter: 4.3 Finding Directions

Back to question 10

Section D: Long answer question

11 [5] hard

Kabir has an iron sewing needle, a bar magnet, a small piece of cork, a bowl and some water.
(a) Write the steps he should follow to make a magnetic compass. (3 marks)
(b) How can he check that the needle has become a magnet? (1 mark)
(c) He turns the cork gently a few times. What should he see each time if his compass works? (1 mark)

Answer (a) Magnetise the needle by moving a magnet along it many times, pass it through the cork and float the cork on water. (b) It should attract iron filings or steel pins. (c) The needle should come to rest pointing in the same direction every time.

1. (a) Step 1. Put the needle on a table. Move one pole of the magnet along the needle from one end to the other, then lift the magnet.
2. (a) Step 2. Start again from the same end with the same pole. Repeat 30 to 40 times.
3. (a) Step 3. Pass the needle through the cork and float the cork in the bowl of water, with the needle above the water.
4. (b) Bring iron filings or steel pins near the needle. If they stick to it, the needle is a magnet.
5. (c) Each time the cork stops, the needle points along the same line, the north-south direction.

Read in the chapter: 4.3 Finding Directions

Back to question 11

Section E: Case-based question

12 [4] hard

Read the passage and answer the questions.
Four ring magnets P, Q, R and T are put on a wooden rod, one above the other. P is at the bottom. Q floats above P without touching it. R sits on Q and touches it. T floats above R without touching it. The top face of P is a North pole.

(i) Which pole is the bottom face of Q? Give a reason.
Answer North. Q floats, so it is repelled by P. The faces that face each other must be like poles.

(ii) Which pole is the top face of Q?
Answer South. It is the other pole of the same magnet.

(iii) Which pole is the bottom face of R? Give a reason.
Answer North. R touches Q, so it is attracted. The faces that touch must be unlike poles, and the top of Q is South.

(iv) Which pole is the bottom face of T?
Answer South. The top face of R is South, and T floats, so the facing poles are like poles.

1. Each part carries 1 mark.
2. Rule used: like poles repel (the magnet floats) and unlike poles attract (the magnets touch).

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

Back to question 12